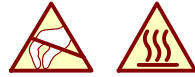
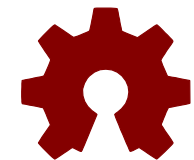
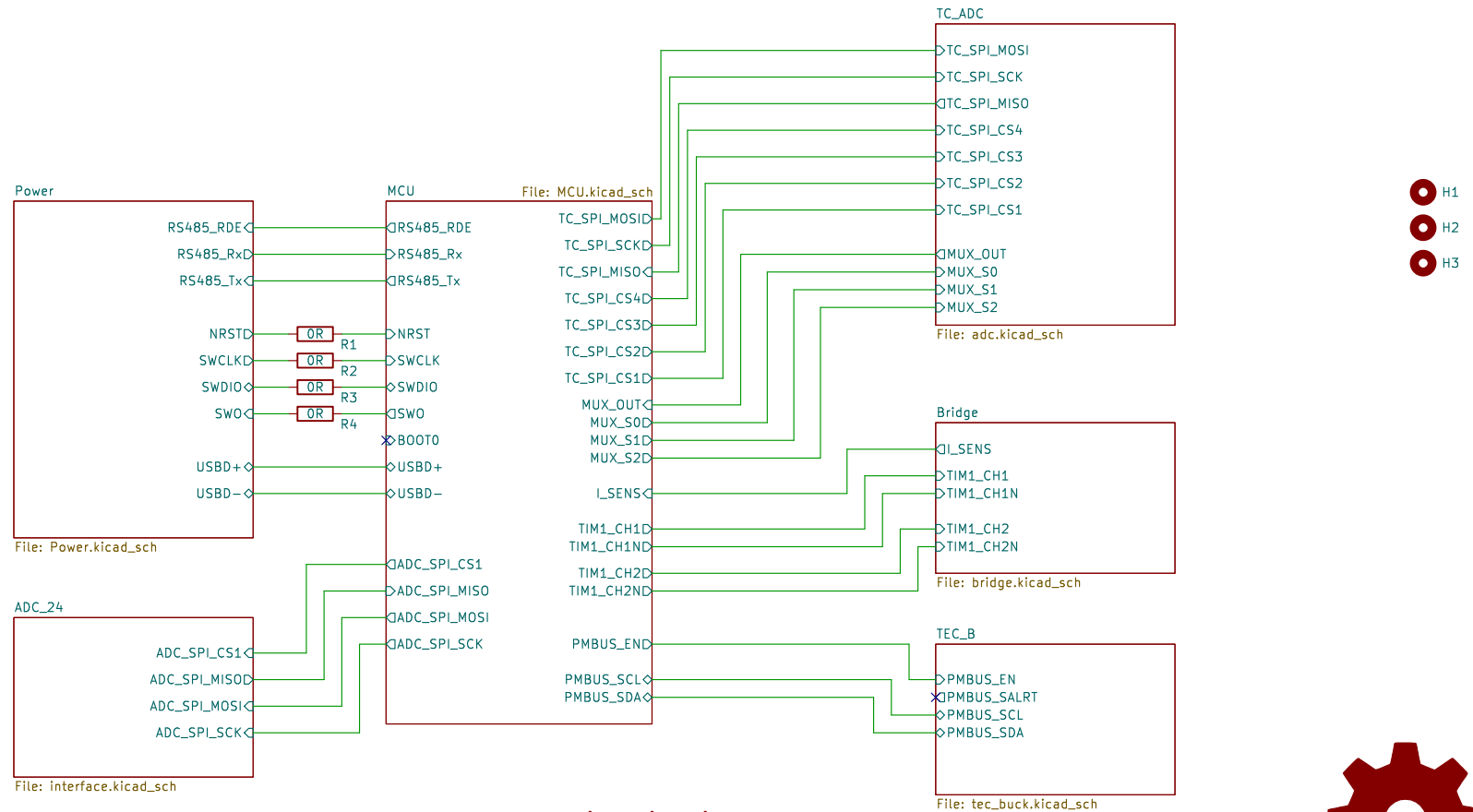


ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC



STM32 6 Channel Precision Thermocouple Interface with isolated RS 485
and 2 Channel (Bidirectional, Unidirectional) Isolated TEC Driver ~240W/120W



UNDER REVISION
DO NOT MANUFACTURE
NOT FOR MEDICAL OR FOR USE IN
CRITICAL APPLICATIONS
OPEN TO PUBLIC
NOT RELATED TO ANY
ORGANIZATION
USE AT YOUR OWN RISK

Simple PC-Controlled
Multi Channel Universal
Thermocouple Interface
-Test TEC Control Methods
-Data analysis
-Isolated RS-485
-Software Controlled
-Precise, low distortion
-IO protection

This version does
NOT feature fully
isolated TC inputs,
or USB isolation.
See version 1.0.2.
Beware of input
devices, ensure
probes are not
externally energized!!!

OPEN TO PUBLIC

ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC

Sheet: /
File: STM32_MULTI_THERMO.kicad_sch

Title: Root

Size: A4

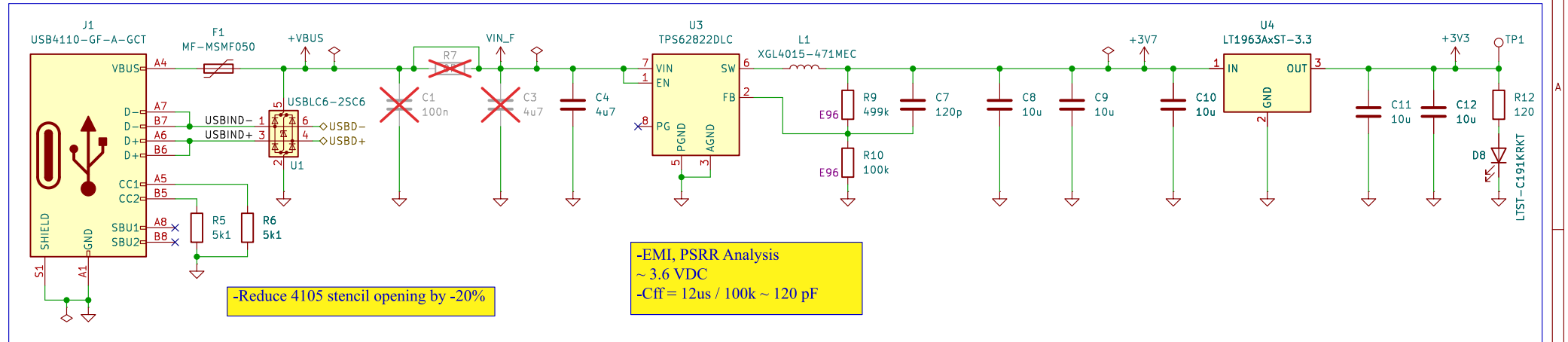
Date: 2026-03-20

Rev: 1.0.0

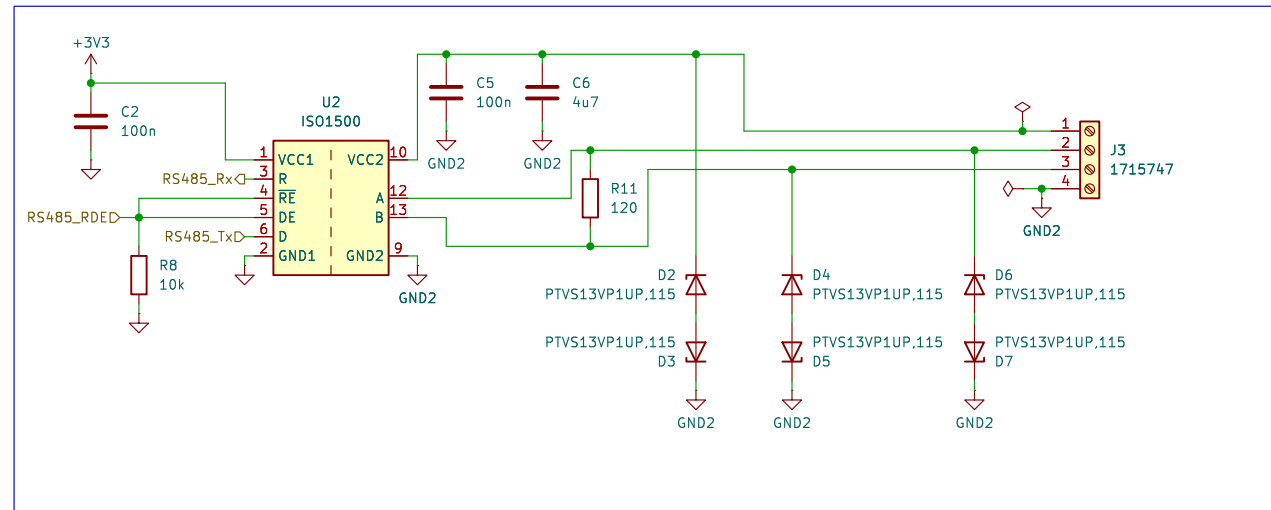
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Id: 1/7

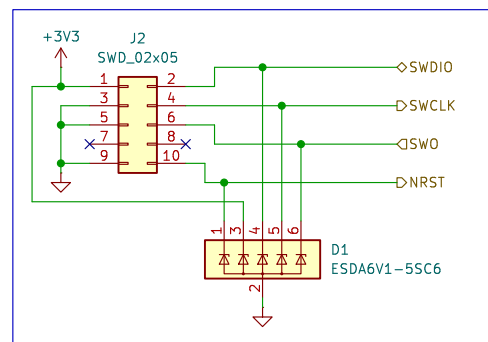
USB-C FILTER, BUCK CONV, LDO STAGE



RS-485 ISO



SWD CONN



ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC

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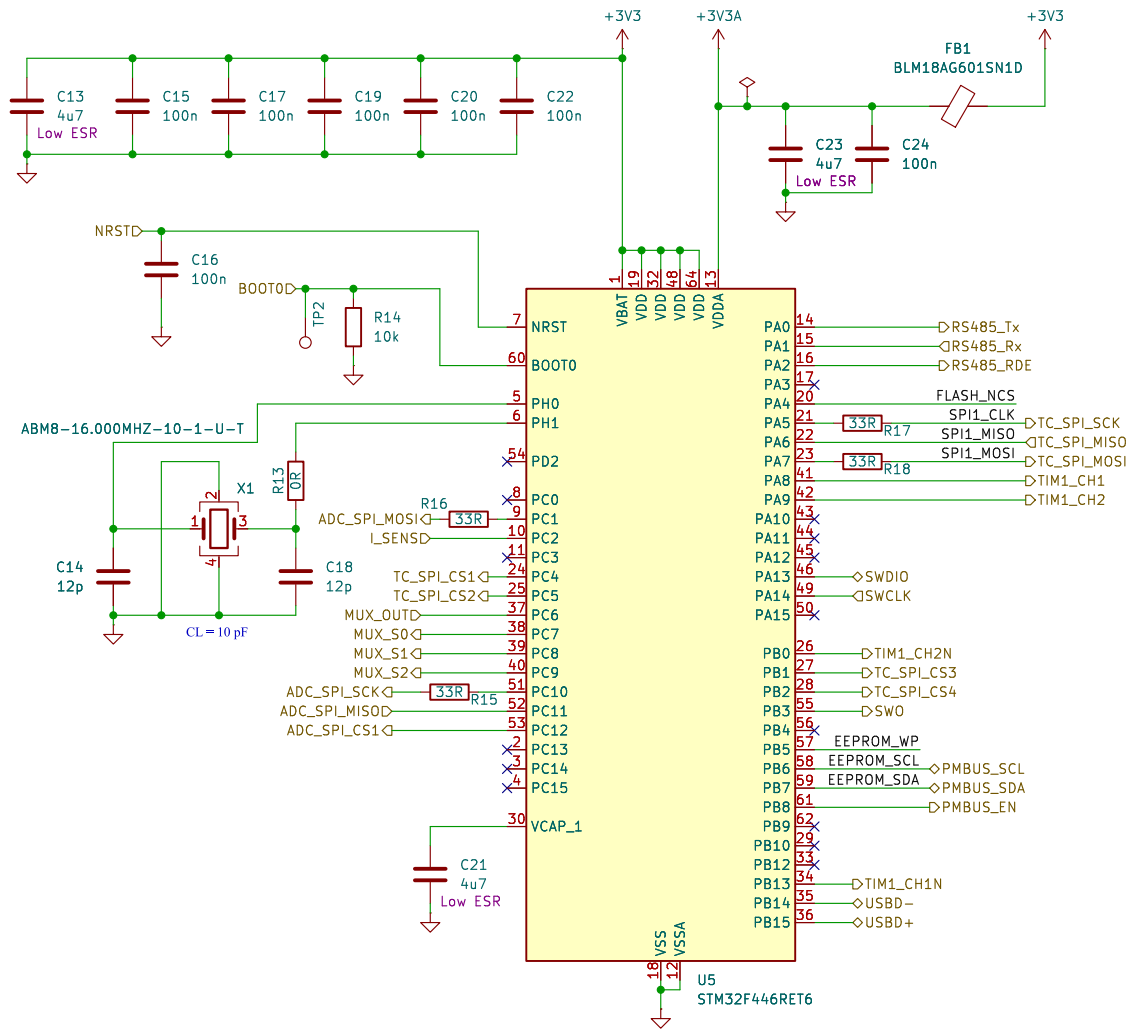
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KiCad E.D.A. 9.0.7

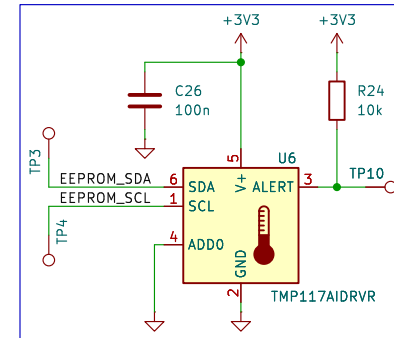
Rev: 1.0.0

Id: 2/7

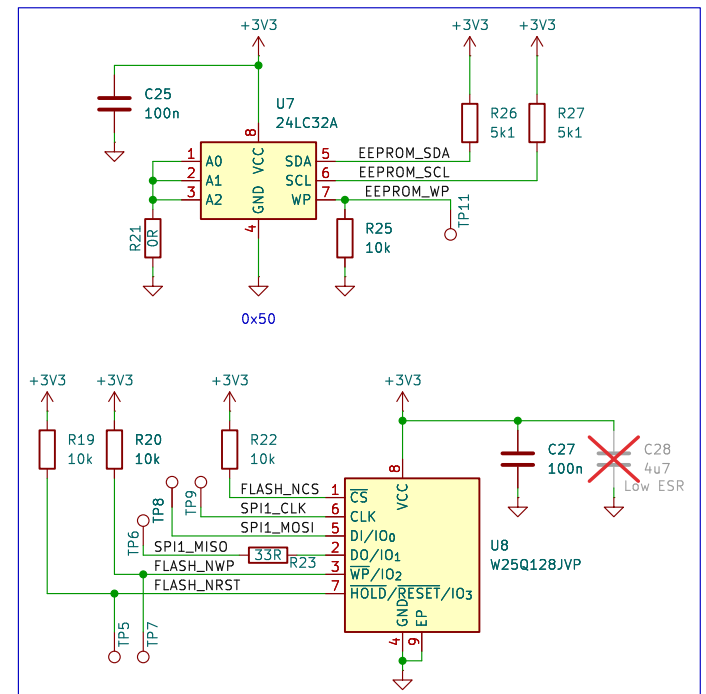


-NEED 22-50R SERIES RESISTORS ON ALL SPI LINES
place near drivers 5x for the spi bus that shares with w25 and max31856s

TEMP REFERENCE



CONFIG MEMORY + MASS STORAGE



ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU

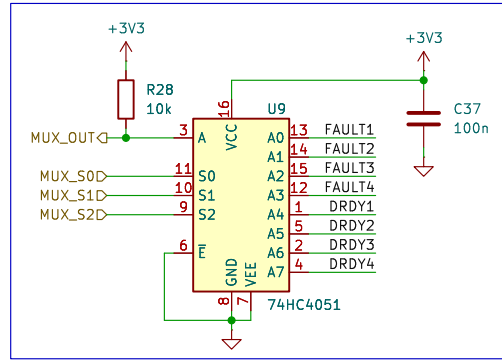
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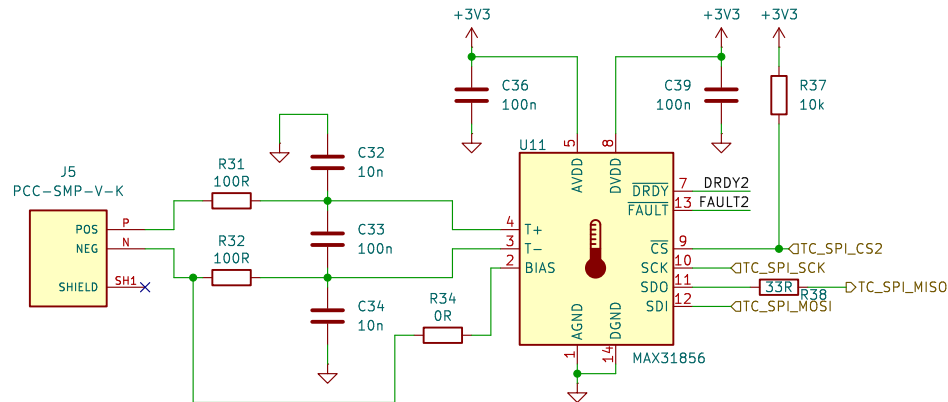
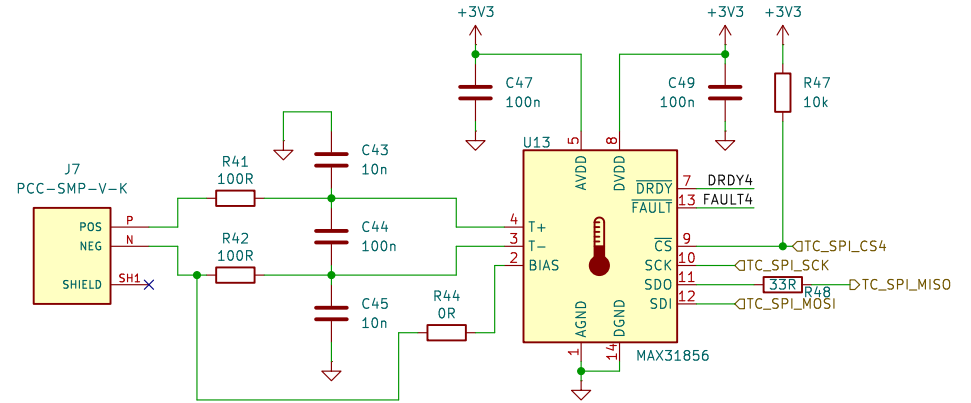
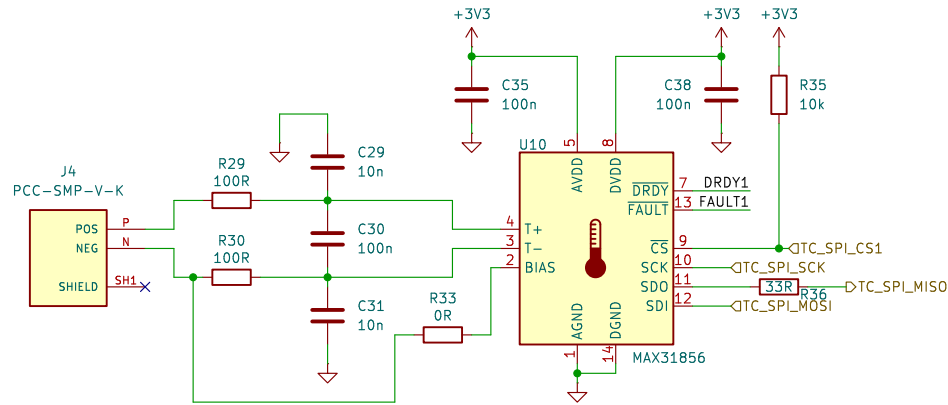
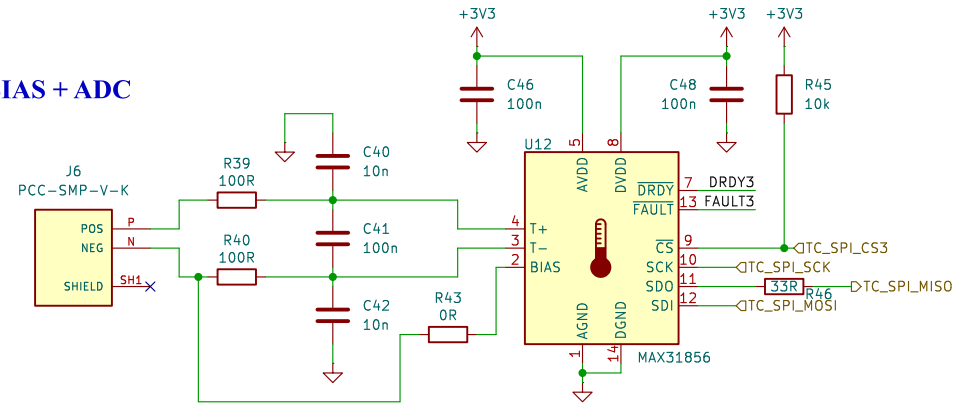
Rev: 1.0.0

Id: 3/7

DRDY/FAULT MUX



TC INPUTS + BIAS + ADC



ISOTHERMAL BLOCK

ENGINEERING PROTOTYPE V1.0.0
OPEN TO PUBLIC

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File: adc.kicad_sch

Title: TC

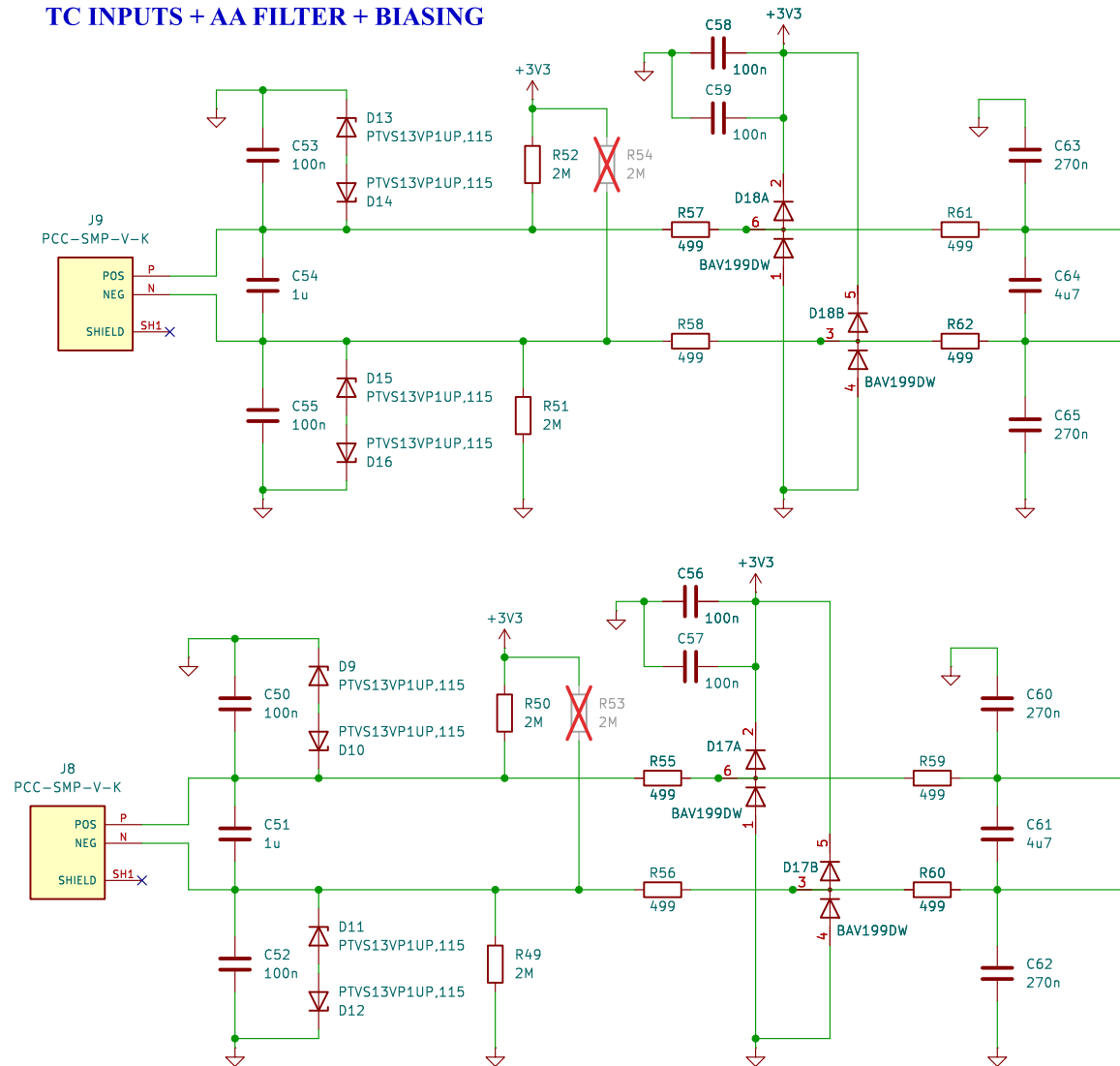
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KiCad E.D.A. 9.0.7

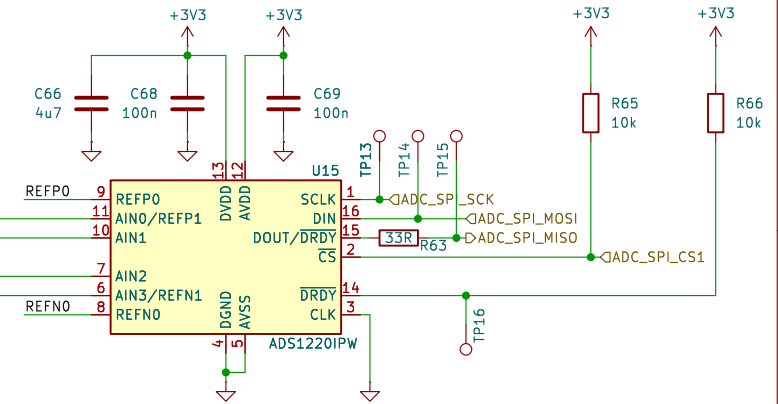
Rev: 1.0.0

Id: 4/7

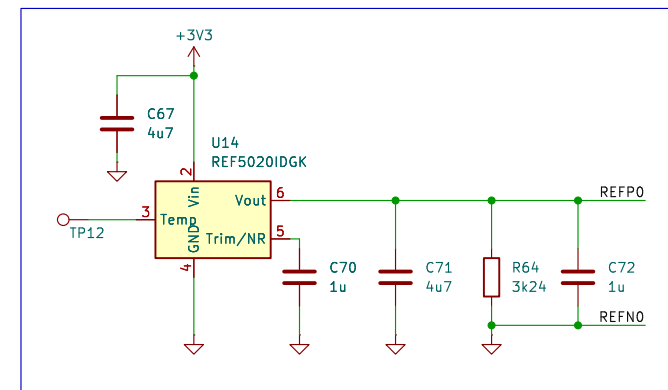
TC INPUTS + AA FILTER + BIASING



ADC



REFERENCE



ISOTHERMAL BLOCK

-ADC FILTER REF TIDA-00168
<https://www.ti.com/lit/ug/tidu574/tidu574.pdf>

INTERNAL TEMP SENSORS ARE
 USED FOR CJC. CAN LEAVE
 CONNECTOR MECHANICAL ASSEMBLY
 FLOATING!!!!

ENGINEERING PROTOTYPE V1.0.0
 OPEN TO PUBLIC

Sheet: /ADC_24/
 File: interface.kicad_sch

Title: ADC_24

Size: A4 Date: 2026-03-23
 KiCad E.D.A. 9.0.7

Rev: 1.0.0
 Id: 5/7

